Applications in AI and Data Science

Coursework Report: Property Price Analytics and Facial Expression Detection

COMP1891 (2025/2026)

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1 Section A: Data Science (Scenario 1)

1.1 A1 – Exploratory Data Analysis

The supplied file `Estate_Agent.csv` contains 350 records and nine columns. It is semicolon-delimited rather than comma-delimited, so the loader detects the separator from the header line instead of assuming a default; read with the wrong separator the file collapses into a single unusable column.

The nine columns divide into three roles. `House_Price` is the regression target. `Rating` is an ordinal quality grade used as the classification target in Section B. The remaining seven – `House_Area`, `Build` (year of construction), `No_of_Bathrooms`, `No_of_Bedrooms`, `Garage_Area`, `Living_Space` and `Parking` – are predictors describing the physical property.

All seven predictors are retained, on two tests: does the feature describe the property, and does it carry signal? Table 1 shows every predictor correlating with price at $|r| \geq 0.15$, and none is a proxy for a protected characteristic – there is no postcode, neighbourhood or demographic column, which materially reduces the discrimination risk discussed in Section D. `Rating` is deliberately excluded: it correlates with price at $r = -0.045$, and is a subjective assessment rather than a physical measurement, so including it would make the model depend on a human judgement the agency wants to benchmark.

Table 1: Pearson correlation with `House_Price` (after cleaning).

Feature	***r* with House_Price**
Parking	0.691
Garage_Area	0.680
Living_Space	0.608
No_of_Bathrooms	0.531
Build	0.531
House_Area	0.286
No_of_Bedrooms	0.156
Rating	-0.045

The strongest predictor is `Parking` ($r = 0.691$), narrowly ahead of `Garage_Area` ($r = 0.680$). Both measure the same underlying attribute – garage capacity as a count and as an area – and are strongly correlated with each other, which matters for interpreting the regression coefficients in Section B1.

The most commercially significant finding is the near-zero correlation between `Rating` and price: the agency's own quality grade does not predict what a property sells for. Either the grade captures something buyers do not pay for, or it is applied inconsistently, and both readings recommend reviewing the grading process before relying on it for pricing strategy. Figure 1 presents the full correlation matrix.

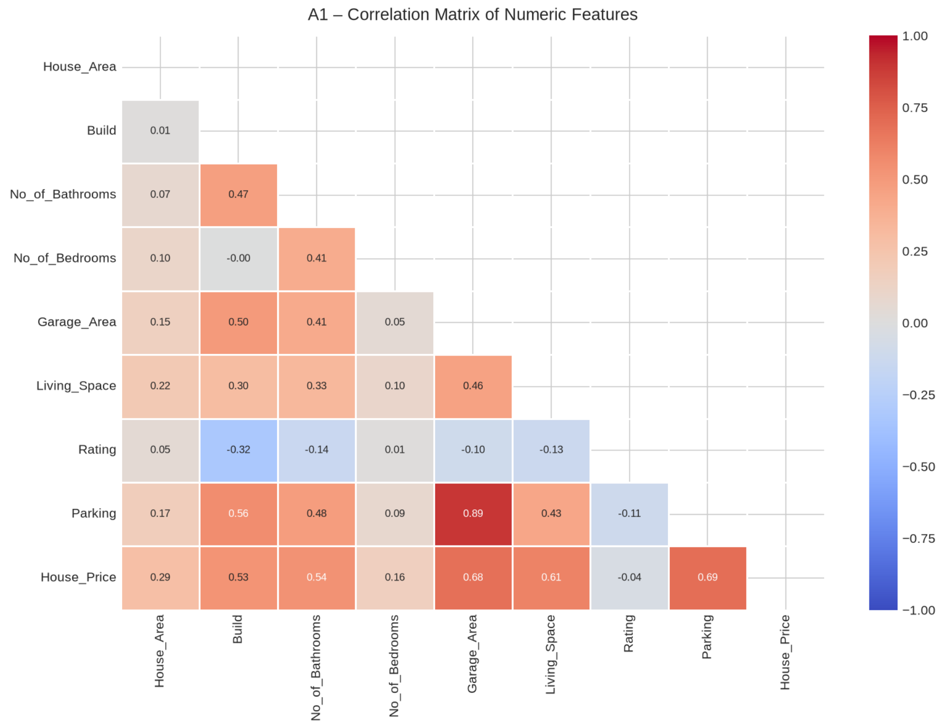

Figure 1: A1 – Correlation matrix of all numeric features.

1.2 A2 – Data Cleaning

The raw data is not analytics-ready. Four distinct defects were found and corrected.

Formatting errors in a numeric column. `Parking` is stored as text and mixes digits with written words: nine records hold `One` (4), `Two` (3) or `Three` (2). A naive conversion turns all nine into missing values, which median imputation then silently overwrites with plausible-looking numbers. Since `Parking` is the strongest price predictor, this is the most damaging defect in the file. The cleaning routine maps written numbers to integers before conversion and reports what it changed.

Duplicate records. Four exact duplicates were removed (350 $\rightarrow$ 346). They bias every statistic and can leak the same property into both training and test sets.

Missing values. Four columns had gaps: `Living_Space` (5), `Garage_Area` (3), `No_of_Bathrooms` (2) and `No_of_Bedrooms` (2), never more than 1.4% of any column. Median imputation was used rather than mean because the area columns are right-skewed, and the median is not dragged by the upper tail.

Outliers. Treatment was matched to each column rather than applied as one blanket rule (Table 2). Price uses the conventional 1.5$\times$IQR fence, excluding 14 records. The area columns are naturally right-skewed – large plots are genuine, not erroneous – so the same rule would discard 17 valid properties from `House_Area` alone. A 3$\times$IQR fence was used there, removing only the 5 implausible extremes, including one plot of 159,000 square feet against a median of 9,378. The data falls from 346 to 327 rows.

Table 2: Outlier audit. The applied rule is shown in bold.

Column	Fence	Lower	Upper	Outliers	% of rows
House_Price	**1.5 IQR**	688	343,188	**14**	4.0
House_Price	3.0 IQR	−127,750	471,625	2	0.6
House_Area	1.5 IQR	1,603	17,569	17	4.9
House_Area	**3.0 IQR**	−4,384	23,556	**6**	1.7
Living_Space	1.5 IQR	173	2,075	5	1.4
Garage_Area	1.5 IQR	−63	959	3	0.9

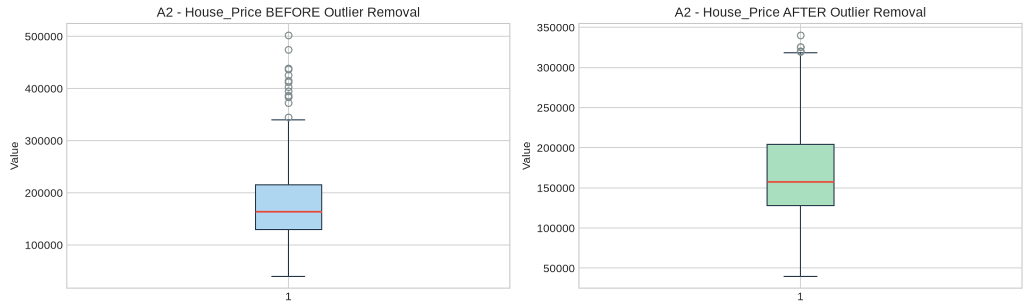

Figure 2: A2 – Price distribution before and after outlier removal.

1.3 A3 – Data Visualisation

Each visualisation was chosen to answer a specific question rather than to display the data generically.

Price distribution (Figure 3). Linear regression assumes normal residuals, and a histogram with a Q–Q plot tests that directly. Post-cleaning skewness is 0.707, below the $|\text{skew}| > 1$ threshold for a log transformation, so price was modelled untransformed.

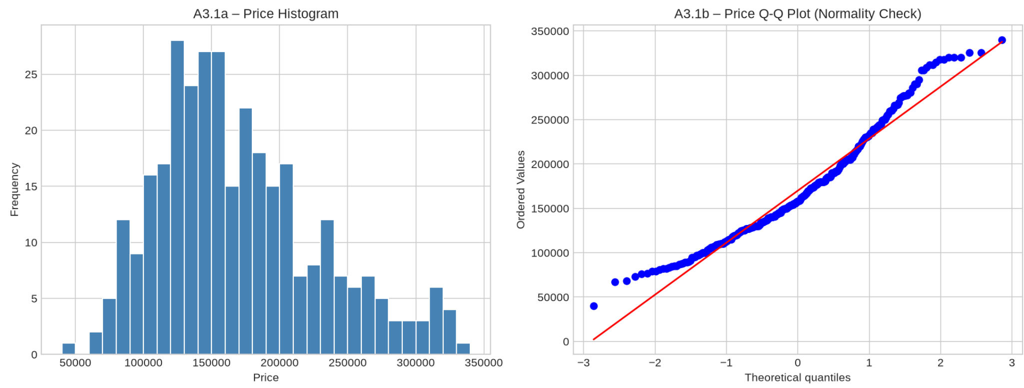

Figure 3: A3.1 – Price histogram and Q–Q plot.

Feature distributions (Figure 4). Count and continuous variables are plotted differently: bedrooms, bathrooms, rating and parking take few whole-number values, so histograms would invent fractional bins and hide category imbalance. They are drawn as category bars annotated with the largest level's share; continuous features keep histograms.

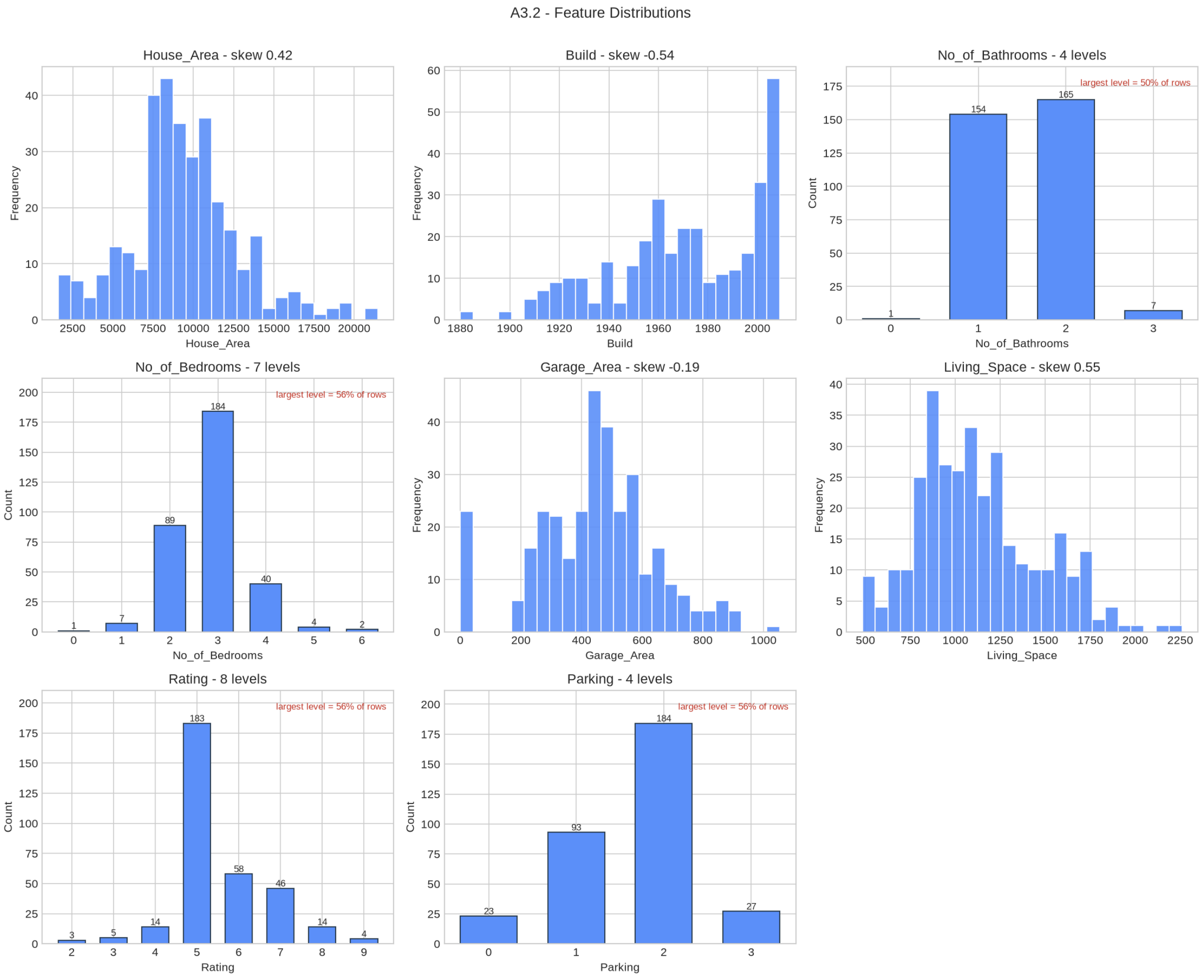

Figure 4: A3.2 – Feature distributions, split by variable type.

Price against the leading predictors (Figure 5). Scatter plots with a fitted trend line test whether the correlation reflects a genuinely linear relationship, which a single coefficient cannot distinguish from a curved one.

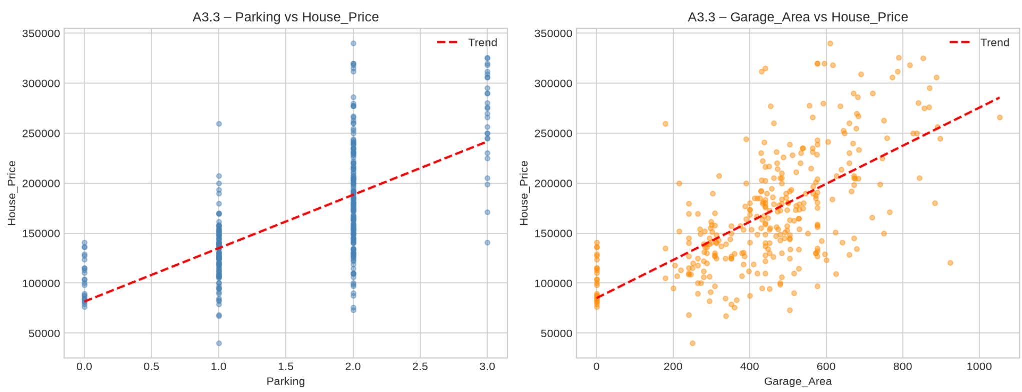

Figure 5: A3.3 – Price against the two most correlated features.

Price by category (Figure 6) shows price dispersion across grades and bedroom counts, with group sizes on the axis so a wide box resting on three properties is not read as a pattern. It confirms visually what Table 1 states: median price does not climb consistently with rating.

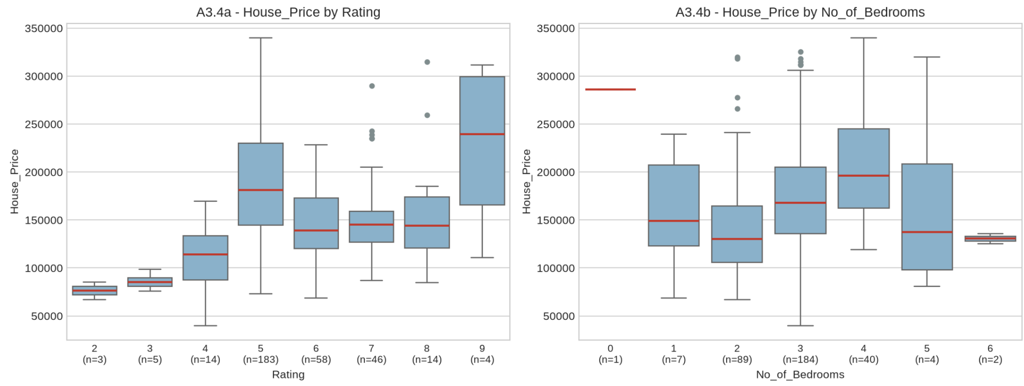

Figure 6: A3.4 – Price by rating and by bedroom count (n shown per group).

Class balance (Figure 7). `Rating` is the Section B target, so its balance was checked before any model was fitted. Grade 5 holds 56.0% of properties, the imbalance ratio is 61:1, and grades 2, 3 and 9 have under ten examples each. This determines the choice of metric throughout Section B: predicting grade 5 unconditionally already scores 56% accuracy while learning nothing.

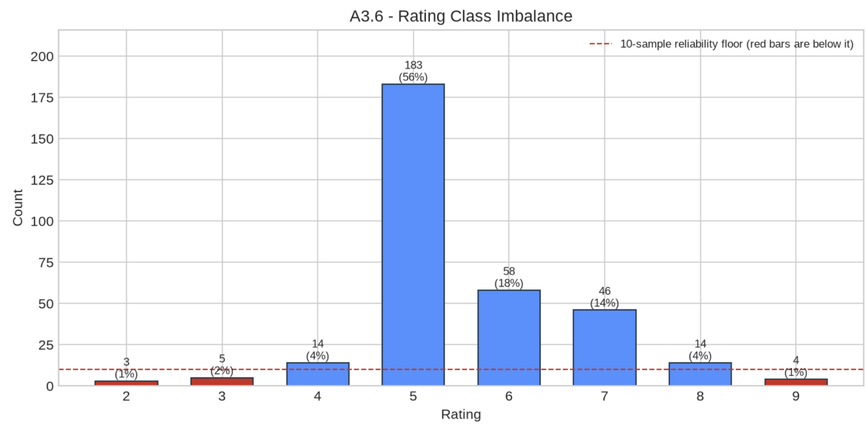

Figure 7: A3.6 – Rating class imbalance. Red bars fall below a usable sample size.

2 Section B: AI and Machine Learning (Scenario 1)

All models use the same seven predictors inside a `Pipeline` of median imputation, standardisation and the estimator. Placing preprocessing in the pipeline matters: fitted once on the full dataset, the scaler and imputer would learn the hold-out rows' statistics and leak them into training. Measured on the banded task of B3.3, that leakage inflated cross-validated macro-F1 from 0.469 to 0.497 and selected a different k. Inside a pipeline both steps refit on each training fold alone.

2.1 B1 – Multiple Linear Regression

Three train/test splits were evaluated (Table 3). Three metrics are reported together because each answers a different question: MAE gives the average error in pounds and is what a client would understand; RMSE penalises large errors more heavily and so exposes occasional bad predictions; R^2 states the proportion of price variance explained.

Table 3: B1 – Linear regression across three train/test splits.

Split	Train	Test	MAE	RMSE	R^2
70/30	228	99	23,701.57	31,967.72	0.763
80/20	261	66	24,113.45	31,915.02	0.763
90/10	294	33	23,760.23	30,314.78	0.810

The reporting split was fixed at 80/20 *before* any result was seen; choosing the best-scoring split afterwards would select a model on the test set and report the flattering 0.810 from the 90/10 split, whose 33-property test set is the least reliable. On the declared split the model explains 76.3% of variance with a typical error of £24,113 against a median price near £164,000, about 15%.

Even that is optimistic. Five-fold cross-validation gives $R^2 = 0.637 \pm 0.148$, folds ranging 0.443 to 0.790. The single-split figures cluster near 0.76 only because they share one convenient partition; the honest estimate is twelve points lower and far less stable. The wide fold spread is itself a finding: with 327 records, performance depends materially on which properties the model happens to train on.

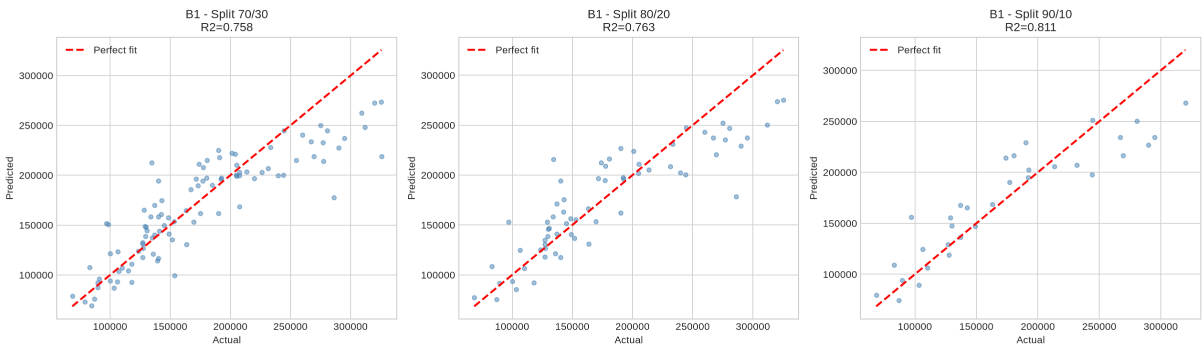

Figure 8: B1 – Predicted against actual price for each split.

Figure 10 shows the coefficients, but they must be read with care. Variance inflation factors confirm the redundancy anticipated in A1: `Parking` reaches VIF = 5.31 and `Garage_Area` 4.87, so 81% and 80% of each is predictable from the other features. Above the conventional threshold of 5 a coefficient is unstable, because the model cannot separate two overlapping measurements of the same garage provision. Predictive accuracy is unaffected – only interpretation is. The other five features sit below VIF = 1.8.

Residual diagnostics (Figure 9) test the underlying assumptions. Residuals average near zero and are close to normal (skew 0.36), so the model is unbiased overall. But corr(fitted, |residual|) = 0.438: error grows with predicted price. This heteroscedasticity makes the model less precise on expensive properties, so the single MAE understates uncertainty at the top of the market. High-value valuations warrant a wider interval; modelling log(price) is the standard remedy.

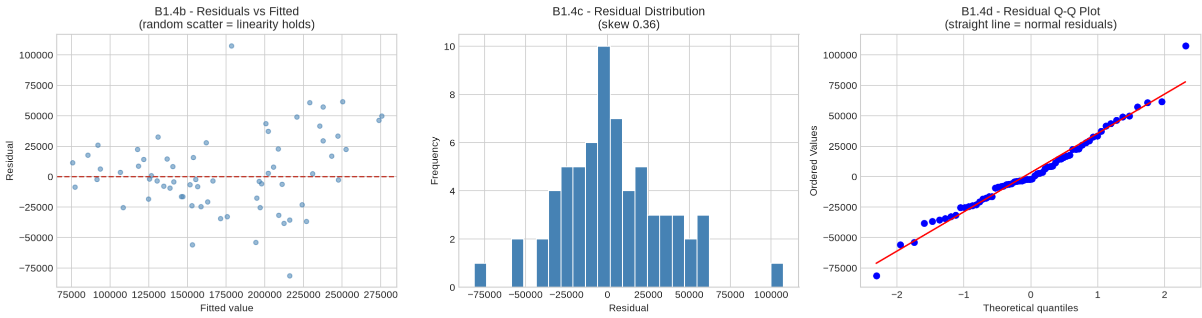

Figure 9: B1 – Residual diagnostics: residuals against fitted values, their distribution, and a normal Q–Q plot.

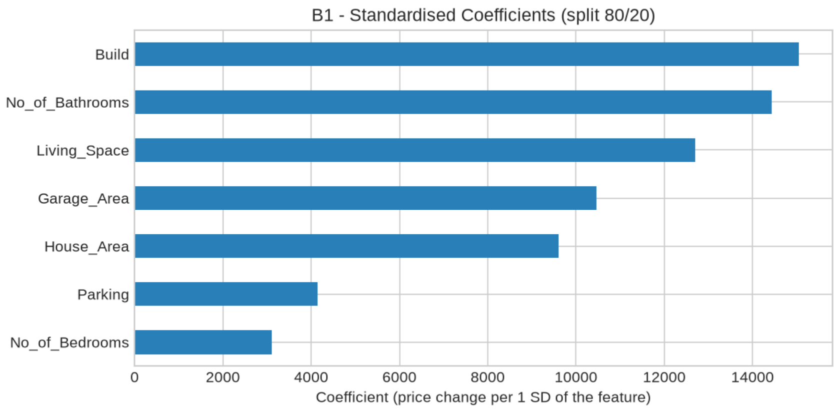

Figure 10: B1 – Standardised feature coefficients.

Applying the model to a median property (9,180 sq ft plot, built 1972, 2 bathrooms, 3 bedrooms, 454 sq ft garage, 1,085 sq ft living space, 2 parking spaces) predicts **£181,951**. Given the MAE, this should be quoted to a client as approximately £182,000 $\pm$ £24,000 rather than as a point figure.

2.2 B2 – k-Nearest Neighbours

k was tuned across $k = 1 \ldots 25$ using stratified cross-validation on the training set only. Selecting k by test-set accuracy – a common shortcut – leaks the test data into model selection and produces an optimistic score. The difference is visible here: cross-validation selects $k = 11$, whereas $k = 7$ maximises test accuracy. Only the former is a legitimate choice.

The extreme imbalance forces a compromise. The smallest training class holds just two properties, capping the cross-validation at two folds – too few for a stable estimate, which is why the selected k is itself unreliable here.

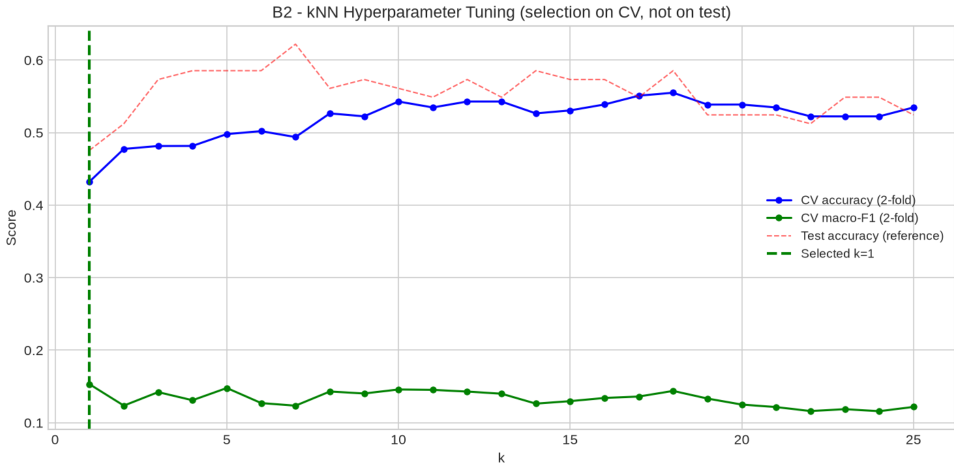

Figure 11: B2 – kNN tuning. Selection uses the cross-validated curve, not the test curve.

Cross-validation selects $k = 1$, which achieves 47.6% accuracy – well *below* the 56.1% majority-class baseline. Balanced accuracy is 0.163 and macro-F1 0.161. That $k = 1$ wins is itself diagnostic: on two folds of a severely imbalanced sample, no value of k separates the grades, so the search settles on the most flexible one. The per-class report confirms it – grades 2, 3, 4, 8 and 9 all score zero on both precision and recall. Reporting only weighted F1 (0.497) would disguise this completely, which is why macro-averaged metrics are shown.

2.3 B3 – Support Vector Machine

Three kernels were tuned over C (and γ) by cross-validation, with `class_weight='balanced'` so rare grades are not discarded in favour of the dominant one (Table 4).

Table 4: B3 – Tuned, class-balanced SVM by kernel.

Kernel	Best params	CV macro-F1	Accuracy	Balanced acc.	Macro-F1
Linear	$C = 1$	0.167	0.476	0.447	0.257
RBF	$C = 1$, γ=scale	0.173	0.512	**0.488**	**0.333**
Poly (3)	$C = 1$, γ=scale	**0.180**	0.512	0.315	0.251

Selection used cross-validated macro-F1, favouring the polynomial kernel (0.180) over RBF (0.173); on the test set the ranking reverses (0.333 against 0.251). With two folds and rare classes the CV estimate is itself noisy, and honouring a pre-declared rule sometimes means reporting the model that did not win on the day.

Against kNN the SVM wins on every imbalance-aware metric: accuracy 0.512 vs 0.476, balanced accuracy 0.315 vs 0.163, macro-F1 0.251 vs 0.161. Since the agency needs to identify unusually high- and low-grade properties rather than restate the most common grade, the SVM is the better model – though neither clears the 0.561 baseline.

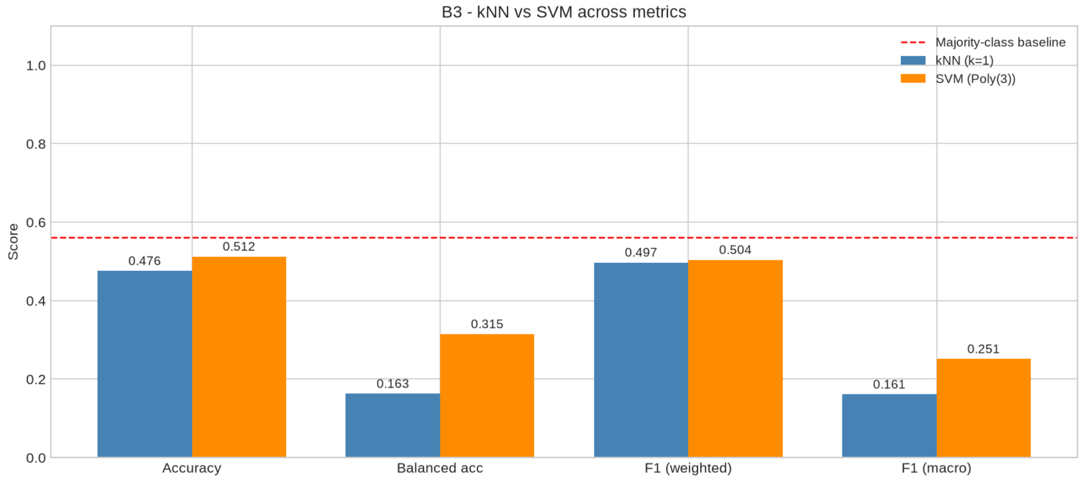

Figure 12: B3 – kNN against SVM across four metrics, with the majority-class baseline.

Addressing the imbalance. Neither model beats the baseline convincingly, because grades with three to five examples cannot be learned. Grouping the eight grades into three commercially meaningful bands – Low (≤ 4, $n = 22$), Mid (5–6, $n = 241$) and High (≥ 7, $n = 64$) – keeps the business question intact while giving every class usable support, and restores full 5-fold cross-validation. Macro-F1 rises from 0.323 to 0.503 (Table 5).

Table 5: B3.3 – Banded rating target (baseline accuracy 0.732).

Model	Accuracy	Balanced acc.	Macro-F1
kNN ($k = 5$)	0.683	0.422	0.436
SVM (RBF, $C = 10$, $\gamma = 0.1$)	0.634	**0.561**	**0.503**

The improvement does not come from a smarter model; it comes from asking the question at a resolution the data can support. An estate agent advised on whether a property is a low, mid or high grade asset is well served. A claim to separate grade 8 from grade 9 on three examples is not defensible.

2.4 B4 – Clustering and Dimensionality Reduction

k-means was run for $k = 2 \ldots 10$ and evaluated by the elbow method and silhouette score. The silhouette peaks at $k = 2$ (0.287), giving two balanced segments of 161 and 166 properties – broadly a smaller/older and a larger/newer segment.

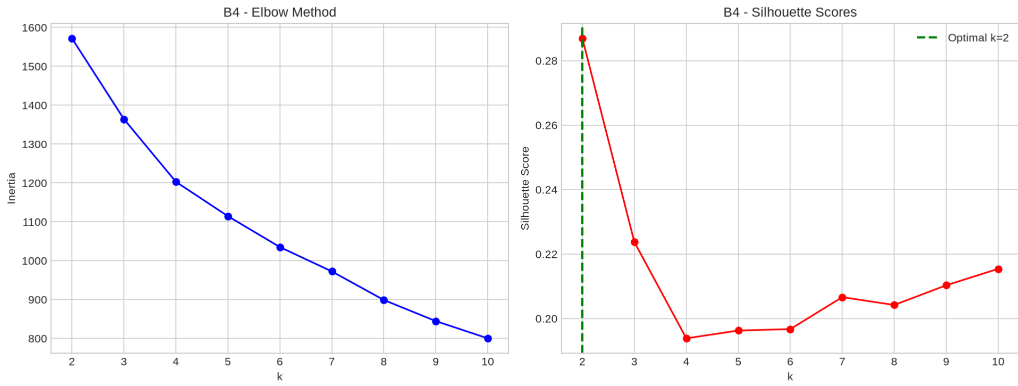

Figure 13: B4 – Elbow and silhouette curves.

Both PCA and t-SNE were applied to project the seven features into two dimensions. Rather than asserting which performs better, both were measured (Table 6).

Table 6: B4 – PCA against t-SNE.

Space	Silhouette	Variance retained	Interpretable axes	Deterministic
Original (7-D)	0.287	100%	yes	yes
PCA (2-D)	0.420	61.6%	yes	yes
t-SNE (2-D)	**0.568**	n/a	no	no

t-SNE produces visually tighter groups (silhouette 0.568 against 0.420), but its axes carry no units, between-cluster distances are not meaningful, and the layout changes with the seed and perplexity. PCA retains 61.6% of variance in two components with axes interpretable as weighted combinations of the original features. PCA is therefore the more appropriate estimator here, because the agency must explain to a client *why* a property falls into a segment; t-SNE remains better for the narrower task of confirming visually that two separable segments exist.

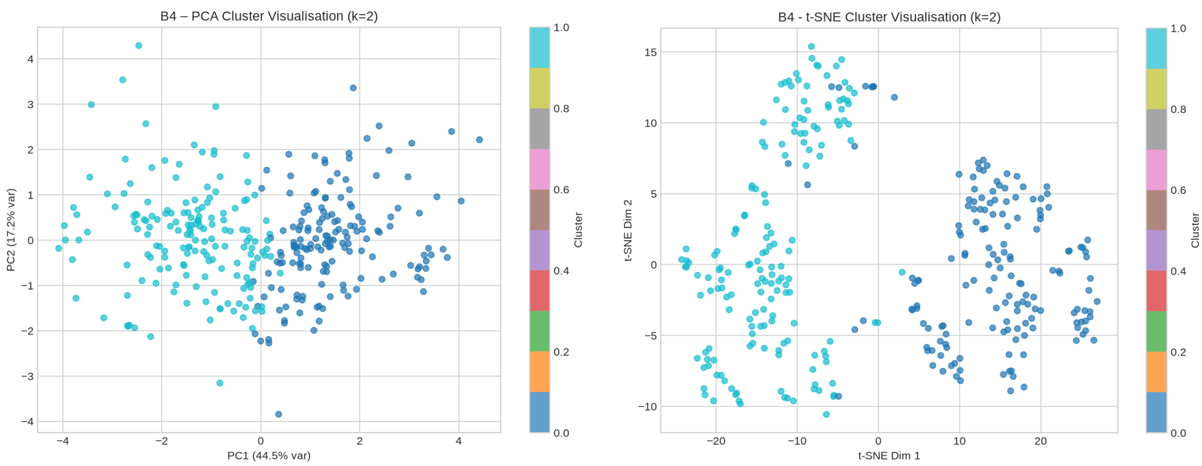

Figure 14: B4 – Clusters under PCA (left) and t-SNE (right).

3 Section C: AI – Facial Expression Detection (Scenario 2)

3.1 C1 – Detection and Measured Accuracy

Fifty images in `FacesSample` were classified into the six expressions named in the brief: Surprised, Angry, Happy, Sad, Frightful and Neutral. An `Expressions` folder holding one self-selected reference image per expression was created as required. Crucially, the sample ships with ground-truth labels (`groundtruth.json`), so detection success is *measured* rather than estimated by eye.

Two methods were implemented and compared. Both use `trpakov/vit-face-expression`, a Vision Transformer fine-tuned on facial expressions.

- **Method 1 – reference matching.** Every image is represented by its 768-dimension CLS embedding, and each sample image takes the label of the most cosine-similar reference image. This uses the `Expressions` folder as the task specifies, and is one-shot learning: a single exemplar per class.
- **Method 2 – direct classification.** The same network's 7-way softmax head predicts the expression directly, having been trained on thousands of labelled faces. Its `disgust` output has no counterpart in the brief and is folded into Angry.

Table 7: C1 – Detection accuracy against ground truth ($n = 50$).

Method	Accuracy	Balanced acc.	Macro-F1
Random guess (6 classes)	0.167	–	–
Majority class	0.260	–	–
Method 1 – reference match	0.660	0.592	0.570
Method 2 – ViT classifier	**0.720**	**0.708**	**0.684**

How successful is the detection? Both beat the 26.0% majority-class baseline, so both genuinely detect expression. Method 2 reaches 72.0%; Method 1 reaches 66.0%, strong for learning each expression from one example, and confirming the embedding space groups faces by expression rather than identity or lighting. The 6-point gap is the cost of one-shot learning: an unrepresentative exemplar harms an entire category. Method 1 scores 0.00 F1 on Frightful, failing all four such faces because its single reference does not resemble them, while Method 2 recovers them at 0.75 recall.

Are any images wrongly detected as Neutral? The methods fail in opposite directions. Method 1 never wrongly assigns Neutral, but misses seven of the nine true Neutral faces – recall of just 0.22. Method 2 wrongly labels two images Neutral, `fig109.jpg` (truly Angry) and `fig145.jpg` (truly Sad), and misses six true Neutrals, for 0.33 recall.

Neutral is the hardest class for both, which is intuitive: defined by the *absence* of expression, it sits close to any weak or partial expression, and a low-intensity angry or sad face is genuinely near-neutral in appearance. This matters for the wellbeing application, where misreading distress as neutral is the error of greatest consequence.

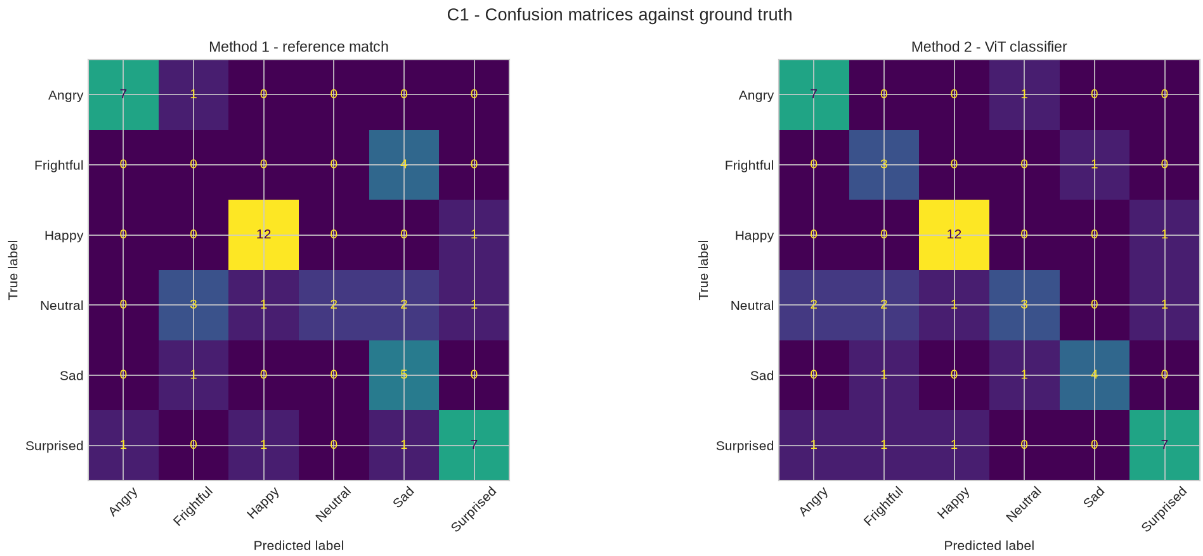

Figure 15: C1 – Confusion matrices for both methods against ground truth.

Prediction confidence is a usable safeguard. Method 2 is 72.9% accurate when its confidence is at least 0.40, but only 50.0% below it. Low-confidence predictions can therefore be routed for human review rather than acted on automatically.

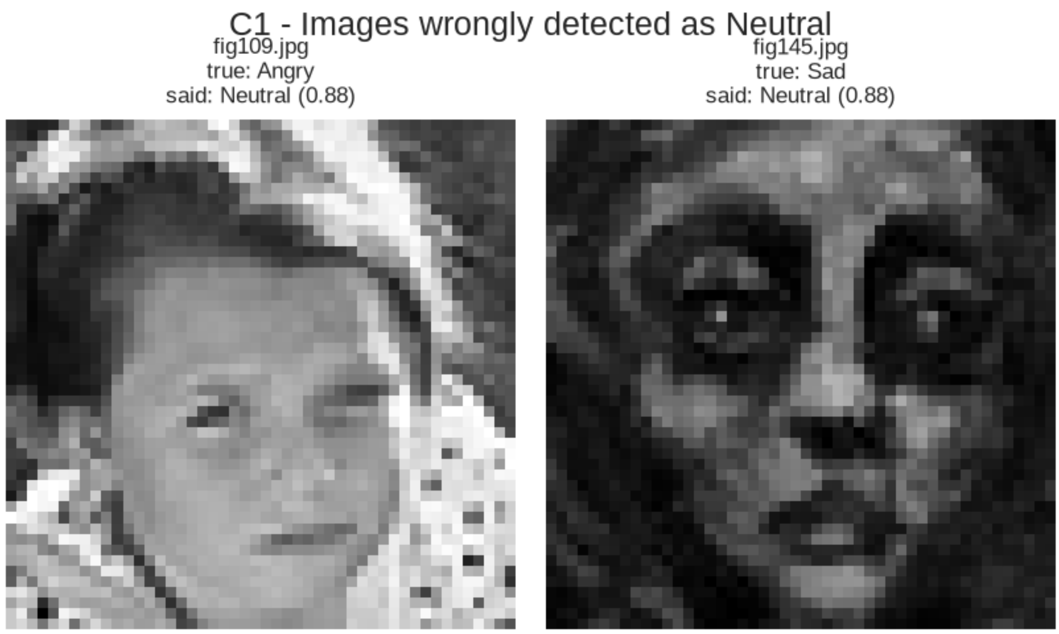

Figure 16: C1 – Images wrongly detected as Neutral.

3.2 C2 – Application: Aggregated Wellbeing Monitoring

C1 turns a face into a label. C2 models the operational system that would sit on top of it for the corporate health and wellness initiative. Expressions are mapped to a stress weight (Happy 0.0 through Angry 0.9), averaged per employee per week over four weeks, and screened by two rules: a *level* rule (weekly mean above 0.55) and a *trend* rule (a rise above 0.15 from the first week to the last).

Two rules are needed because they catch different problems. The level rule finds employees who are consistently under strain; the trend rule catches deterioration early, flagging employees whose absolute score is still moderate but climbing. In the simulation the trend rule flags two employees rising by 0.32 and 0.36 whom the level rule alone would have missed until later.

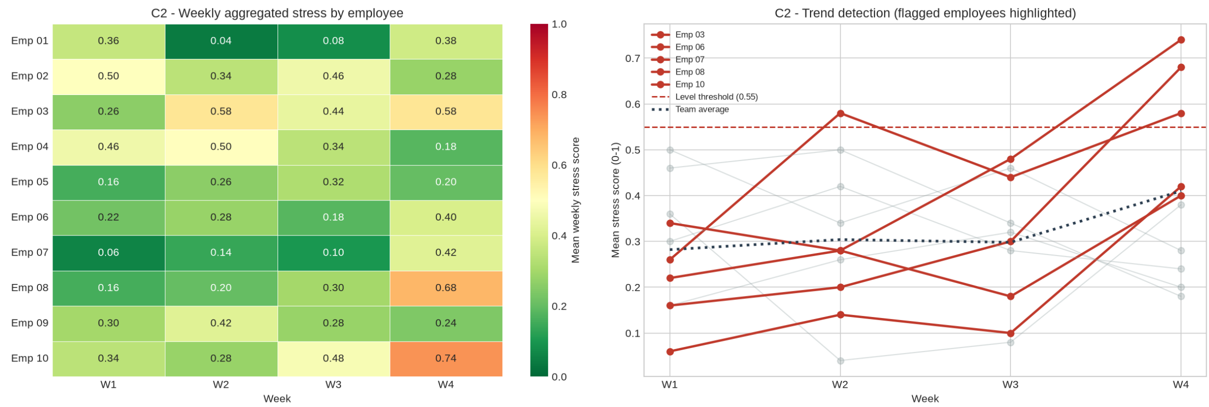

Figure 17: C2 – Weekly aggregated stress and trend detection.

The design is constrained by the ethics discussed in Section D and by the measured accuracy in C1. Rules operate on weekly aggregates, never on individual daily readings; managers see the team average and the number of employees flagged, not the per-person matrix; participation is opt-in and withdrawable. Because C1 measured only 72% accuracy with Neutral the weakest class, a flag prompts an *offer* of a voluntary check-in and never an automated intervention or any input to performance management.

4 Section D: AI and LSEPI

4.1 D1 – Legal, Social, Ethical and Professional Issues

Scenario 1 – price prediction. *Legally*, property records combined with addresses are personal data under the UK GDPR (ICO, 2018); adding postcode – an obvious next step – would introduce a proxy for ethnicity and income, risking indirect discrimination under the Equality Act 2010. *Socially*, models trained on historical sales entrench inequality by under-valuing property in deprived areas, reproducing past disadvantage as prediction (O'Neil, 2016). *Ethically*, they inherit any bias in the human valuations they learn from. *Professionally*, quoting the B1 prediction without its error margin would misrepresent its precision, contrary to RICS transparency obligations (RICS, 2021).

Scenario 2 – expression monitoring. *Legally*, facial images processed to infer mental state are biometric data engaging Article 9 of the UK GDPR, requiring explicit consent and a Data Protection Impact Assessment (ICO, 2011). *Socially*, continuous monitoring erodes psychological safety, and expression norms vary across cultures and with neurodivergence, so a uniform model misreads some groups more than others. *Ethically*, employee consent may not be freely given, since refusal carries perceived career risk (Floridi *et al.*, 2018). *Professionally*, presenting a 72%-accurate classifier – one that misread two distressed faces as Neutral in 50 images – as reliable would breach the ACM requirement to assess and disclose harms (ACM, 2018).

4.2 D2 – Countermeasures

For Scenario 1: complete a DPIA before deployment; exclude postcode and other protected-characteristic proxies, as done here; publish prediction intervals with every point estimate; apply explainable-AI methods such as SHAP; and audit periodically for disparate impact.

For Scenario 2: obtain explicit, withdrawable consent with a penalty-free opt-out; report only aggregates, as the C2 design does; keep a human in the loop so a flag triggers a conversation rather than an action; publish the measured accuracy and weakest class to the workforce; delete raw images once scored; and review error rates across demographic groups.

4.3 D3 – Intelligent Systems for the Common Good

In Scenario 1, transparent pricing models can democratise market information, giving first-time buyers analytical capability once confined to institutional investors, while aggregated signals help local authorities identify gentrification and target affordable housing. The Rating finding shows the wider value: data science can reveal that an established practice does not track the outcome it claims to.

In Scenario 2, the same technology deployed with consent supports early detection of mental health deterioration in clinical settings, and in education can identify distressed students for pastoral support (D'Mello *et al.*, 2017). What separates beneficial from harmful use is not the algorithm but who holds the output, whether participation is voluntary, and whether the error rate is disclosed.

5 Section E: Conclusion and Personal Reflections

5.1 E1 – Conclusion

In Scenario 1, cleaning was where most of the value lay: nine `Parking` values stored as words would have been silently destroyed by a naive numeric conversion, and since `Parking` is the strongest price predictor ($r = 0.691$), that defect alone would have degraded every downstream model. Regression explained 76.3% of price variance on the pre-declared split, but five-fold cross-validation put the honest figure at 0.637 ± 0.148: indicative guidance, not a basis for setting an asking price.

Classifying Rating exposed the limits of the data. With grade 5 holding 56% of records and three grades holding under ten examples, neither kNN (0.476) nor SVM (0.512) beat the 0.561 baseline. Re-posing the task as three bands raised macro-F1 from 0.251 to 0.503 – matching the question to the resolution the data supports mattered more than the choice of algorithm. Clustering found two market segments; t-SNE separated them more attractively, but PCA was selected because only its axes can be explained to a client.

In Scenario 2, both methods beat chance: reference matching against a single exemplar reached 66.0% accuracy, the direct classifier 72.0%. Neutral was weakest for both, and they failed in opposite directions – reference matching never issued a false Neutral but missed seven of nine true ones, while the classifier misread two distressed faces as Neutral. That error shaped the C2 design, where aggregate reporting, dual level-and-trend rules and mandatory human review follow from measured limitations rather than generic caution.

The overriding lesson is that the headline metric is rarely the honest one. Accuracy flattered models that had learned only to repeat the majority class; a single split overstated R^2 by twelve points; one MAE hid that error grows with price; and a bar chart of predicted expressions would have looked plausible while concealing seven missed Neutral faces.

5.2 E2 – Personal Reflections

The most instructive problems here were not the modelling steps but the points where code ran successfully and produced wrong answers.

The clearest was the `Parking` column. My first cleaning routine converted it to numeric without error and reported no missing values afterwards. Only when I printed what the conversion had *changed* did I find that nine values reading `One`, `Two` and `Three` had become NaN and then been overwritten by the median. Nothing failed; the data was simply wrong. Making cleaning routines report what they alter is now a habit, and it caught a second defect – the column detector searched for "year" and never matched `Build`, so year of construction was silently dropped from the feature set.

Leakage taught the second lesson, twice. My initial kNN chose k by test-set accuracy, which I did not recognise as leakage until cross-validation disagreed with it. Having fixed that, I found a subtler instance still in place: the `StandardScaler` was fitted on the whole dataset before splitting, so every model had already seen the hold-out rows' mean and variance. Moving preprocessing inside a `Pipeline` changed both the scores and the selected hyperparameters. Leakage is not one mistake to check off but a category to design against, which is what pipelines exist for. Relatedly, I first judged the classifiers on accuracy and concluded kNN was better, until macro-F1 showed it scoring 0.00 on five of the eight grades: choosing a metric is a modelling decision, not a reporting formality.

A third came from my own C2 simulation. I wrote the expression-probability weights as a positional list, but the labels were sorted alphabetically, so the "calm" weight landed on Angry and the stress model ran backwards – employees scripted to deteriorate appeared to improve, with entirely plausible-looking output. Keying the weights by name removed the whole class of bug, teaching me to prefer constructs in which the mistake cannot be expressed.

Technically I gained fluency in pandas, in scikit-learn pipelines, and in applying pre-trained transformers through Hugging Face, including extracting intermediate embeddings for one-shot matching rather than only using the classification head. Dependency management proved as consequential as modelling: DeepFace required TensorFlow, which had no build for my Python version, so I moved to a PyTorch-only Vision Transformer. Above all I learned scepticism about my own results, and the habit I am taking forward is checking every plausible answer against a baseline – because a model producing sensible-looking numbers while learning nothing is more dangerous than one that visibly fails.

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